

Epitaxial PLZT Mach–Zehnder Modulators with Low $V_{\pi} \cdot L$, Low Loss, and High-Speed Operation

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Abstract Epitaxial PLZT Mach–Zehnder modulators achieve $V_{\pi} \cdot L$ of 0.21 V·cm with propagation loss of 0.8–0.9 dB/cm. Small-signal measurements show ~25 GHz bandwidth, extendable beyond 40 GHz. The devices also support high optical power operation up to 30 dBm. ©2026 The Author(s)

Introduction

The rapid growth of optical access networks and data-center interconnects has intensified demand for optical modulators that simultaneously achieve low driving voltage, low optical loss, and high optical power handling. Reducing modulator power consumption is particularly critical for massively parallel photonic systems, where electrical driving power and thermal management increasingly dominate system-level efficiency.

Thin-film LiNbO₃ (TFLN) modulators offer high bandwidth but suffer from trade-offs in efficiency and high-power operation [1-4]. BaTiO₃ (BTO)-based modulators offer high electro-optic coefficients [5,6] but face challenges related to thin-film uniformity, optical loss, and device reproducibility. Previously reported polycrystalline PLZT modulators [7], while attractive in principle, have been constrained by excess propagation loss and reduced effective electro-optic response.

In this work, we demonstrate epitaxial PLZT ((Pb,La)(Zr,Ti)O₃) Mach–Zehnder optical modulators that overcome these limitations by simultaneously achieving ultra-high modulation efficiency, low optical loss, and robust high optical power operation.

Epitaxial PLZT Platform and Device Design

High-quality epitaxial PLZT thin films were grown on sapphire substrates using a solid-phase epitaxial crystallization process developed for nano-second speed optical switch applications [8-10]. This approach enables stoichiometric control, large-area uniformity, and heteroepitaxial growth while maintaining compatibility with waveguide fabrication processes. X-ray diffraction and pole-figure analysis confirm single out-of-plane and in-plane orientation of PLZT (110) films on sapphire r-plane substrates as shown in Fig. 1, indicating the absence of misoriented grains.

The epitaxial PLZT films exhibit atomically smooth surfaces with sub-nanometer roughness, enabling low-loss optical waveguides after patterning as shown in Fig. 2. Channel waveguides were formed by dry etching, and lateral-field push–pull Mach–Zehnder modulator structures

were implemented with electrode lengths of 1 mm and 2 mm as shown in Fig. 3. This configuration allows efficient phase modulation while minimizing parasitic capacitance and enabling scalable electrode design.

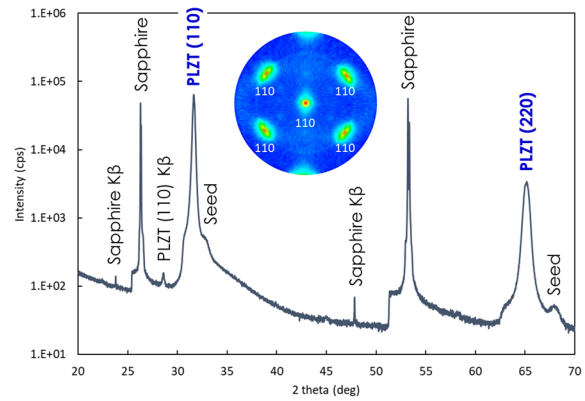


Fig. 1: XRD pattern of PLZT thin film on r-plane sapphire with (110) pole-figure inset confirming in-plane orientation of the PLZT film.

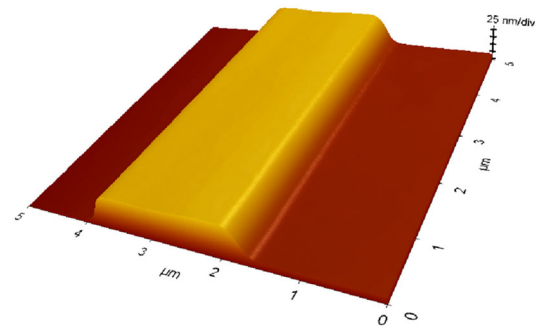


Fig. 2: AFM of etched PLZT waveguide.

Propagation Loss and High Optical Power Handling

Propagation loss was evaluated using cutback measurements on straight epitaxial PLZT channel waveguides as shown in Fig. 4. A low loss of 0.8-0.9 dB/cm was confirmed at a wavelength of 1550 nm in TE mode, representing a substantial improvement over polycrystalline PLZT and BTO-based waveguides. Fig. 5 shows simulated mode profile (1.1 μm x 6.4 μm) and beam

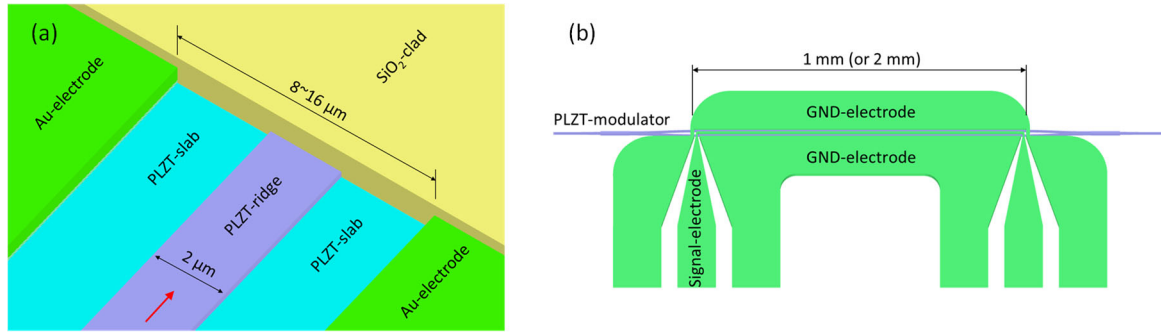


Fig. 3: PLZT Mach-Zehnder modulator (a) cross-sectional view and (b) top view.

propagation coupled from a fiber with mode field diameter of 3.2 μm. The measured coupling loss of 7.2 dB/facet agrees well with the total simulated loss of 7.2 dB/facet, at mode mismatch loss: 6.0 dB, Fresnel loss: 0.2 dB, SMF/small MFD fiber splicing loss: 0.3 dB, and fiber-waveguide gap loss at the distance of 3.0 μm: 0.7 dB, confirming that insertion loss is dominated by coupling rather than propagation loss.

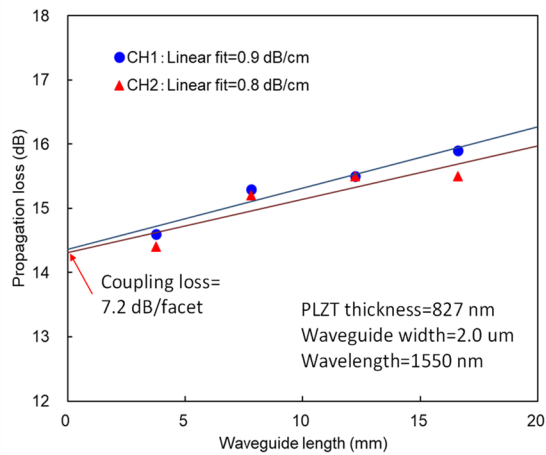


Fig. 4: Propagation loss of epitaxial PLZT waveguides measured by cut-back method, showing low loss of 0.8–0.9 dB/cm at a wavelength of 1550 nm in TE mode.

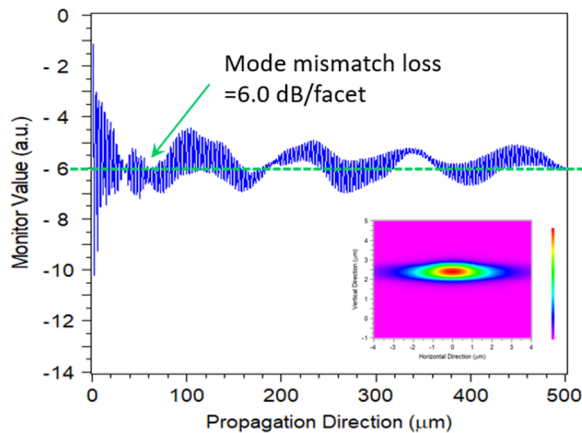


Fig. 5: Simulated beam propagation profile coupled from fiber with MFD of 3.2 μm showing coupling loss components with mode profile inset, confirming that measured insertion loss is dominated by coupling rather than propagation loss.

In addition to low propagation loss, stable operation was confirmed up to an input optical power of 30 dBm (fiber input), corresponding to output power exceeding 20 dBm. The insertion loss in this measurement (~10 dB) differs from that in Fig. 4 due to differences in PLZT waveguide thickness (see Fig. 4) and coupling conditions, and does not reflect the intrinsic propagation loss. No catastrophic degradation or instability was observed up to output power as high as 20 dBm and failure was ultimately caused by degradation of the waveguide–fiber coupling interface, indicating sufficient optical power handling margin such as high-split PON systems.

The low propagation loss demonstrated here, together with robust high optical power handling, establishes epitaxial PLZT as a promising platform for practical integrated modulators. The relatively large fiber-to-chip coupling loss mainly originates from the current facet geometry and coupling scheme, and is not intrinsic to the PLZT waveguide itself.

Electro-Optic Efficiency and Bandwidth Characteristics

The epitaxial PLZT waveguides were poled applying voltage between GND electrodes and electro-optic performance of the epitaxial PLZT modulators was evaluated using DC voltage sweep measurements (Fig. 6 (a)) to extract the intrinsic half-wave voltage V_{π} , independent of high-frequency parasitics and velocity mismatch. For a PLZT thickness of 1.2 μm and a 2-mm electrode length, a minimum V_{π} of 1.05 V was obtained from the 3.65V to 4.70V curve matching, corresponding to an ultra-low reported $V_{\pi} \cdot L$ of 0.21 V·cm as shown in Fig. 6 (b). This $V_{\pi} \cdot L$ is achieved without aggressive driving conditions or favourable metric definitions, but directly reflects the intrinsic electro-optic response of epitaxial PLZT realized in a standard push–pull Mach–Zehnder configuration. Importantly, this performance is reproducible across devices, including both the ultra-low $V_{\pi} \cdot L$ and the low propagation loss, confirming that the results reflect intrinsic device characteristics rather than measurement artifacts. Only reports that explicitly provide both

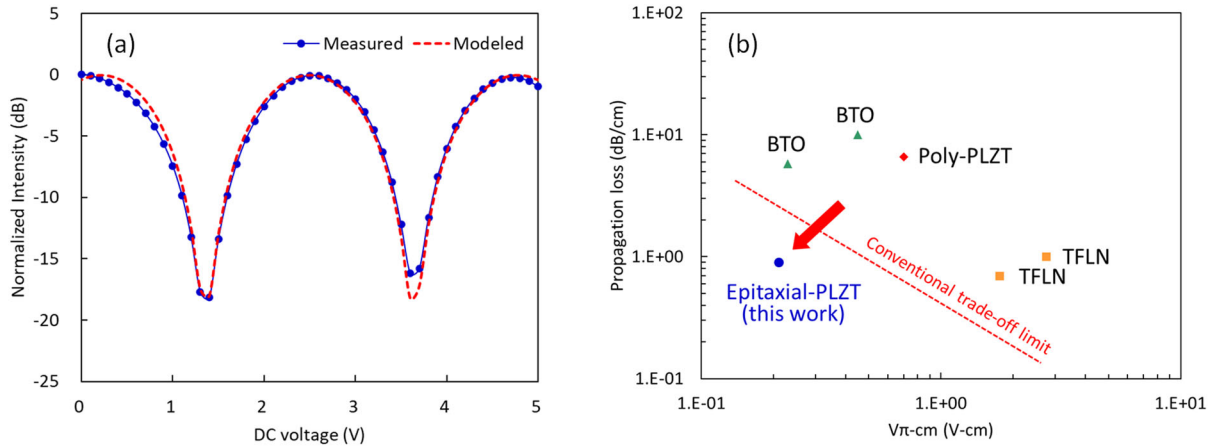


Fig. 6: (a) DC voltage sweep measurements to directly extract the intrinsic half-wave voltage V_π at a wavelength of 1550 nm in TE mode. (b) Propagation loss versus $V_\pi \cdot L$ comparison of integrated modulators. Epitaxial PLZT simultaneously achieves ultra-low $V_\pi \cdot L$ and low propagation loss, surpassing the conventional trade-off limit.

$V_\pi \cdot L$ and waveguide propagation loss under consistent and clearly defined measurement conditions are included in this comparison. To the best of our knowledge, this represents the lowest reported $V_\pi \cdot L$ achieved simultaneously with low propagation loss in an optical modulator.

Electro-optic modelling and fitting yield an effective electro-optic coefficient of 346 pm/V, exceeding values reported for TFLN, BTO, and polycrystalline PLZT integrated modulators. This result highlights the critical role of epitaxial film quality in maximizing the effective electro-optic response at the device level.

The electro-optic frequency response was measured using small-signal modulation, where the bandwidth is defined at the -3 dB point, as shown in Fig. 7 (a). A 3-dB bandwidth of approximately 25 GHz was experimentally verified. Increasing the optical input power significantly reduced the high-frequency noise floor without altering the roll-off behaviour, confirming that the observed bandwidth limitation originates from measurement signal-to-noise ratio rather than

intrinsic device physics.

Separate electromagnetic simulations, as shown in Fig. 7(b), indicate that bandwidths exceeding ~40 GHz are achievable through optimized termination conditions and reduced PLZT loading beneath the electrodes. The frequency response is influenced by impedance mismatch, where moderate termination values provide a balance between bandwidth extension and response flatness. The observed peaking near 54 GHz is attributed to electrode bends.

Conclusion

We demonstrate epitaxial PLZT Mach-Zehnder modulators achieving $V_\pi \cdot L$ of 0.21 V·cm, low propagation loss of 0.8–0.9 dB/cm, and high optical power handling up to 30 dBm. The measured 3-dB bandwidth of ~25 GHz is limited by measurement conditions, while simulations indicate bandwidths exceeding ~40 GHz. These results establish epitaxial PLZT as a highly promising platform for low-voltage, low-loss, and high-output photonic systems.

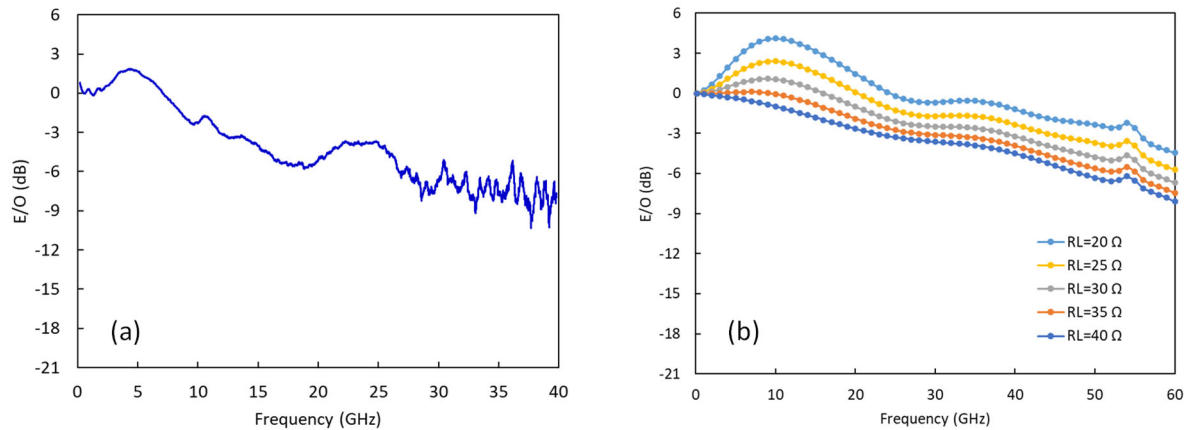


Fig. 7: (a) Measured electro-optic frequency response of an epitaxial PLZT modulator chip obtained by on-wafer RF probe measurements with on-chip termination resistors. (b) Simulated electro-optic frequency response as a function of termination resistance, showing that response shaping is governed by impedance mismatch rather than intrinsic device bandwidth.

Acknowledgements

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